MTH1W

The Rule of Four

You are now able to identify a linear relation from

1. Its **graph**
2. Its **equation**
3. Its **table of values**
4. **Words**

The Rule of 4 states that a relation can be represented in these 4 different ways.

Today: we practice the interconversions of those four representations.

Example 1

The graph below shows the cost of dried apricots.
Apply the **Rule of Four** to the linear relation.

Graph:

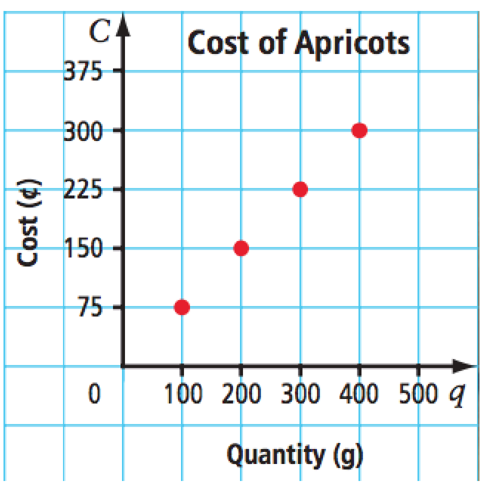


Table of Values

Quantity (g)	Cost ($)

Words

Equation

Example 2

Divers experience an increase in pressure as they dive deeper below sea level.
The table shows the relationship between depth (in meters) and pressure in atmospheres.

Finish the table of values, draw the graph, and give the words + equation for this relationship.

Graph:

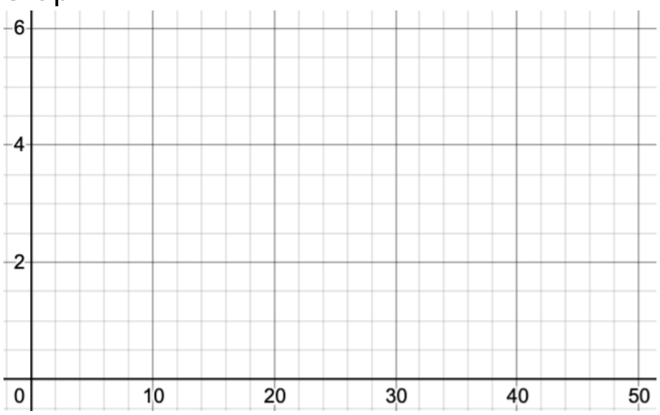


Table of Values

Depth (m)	Pressure (atm)
0	1
10	2
20	3
	4
	5
	6

Words

Equation

Example 3

A particular video club lets members rent vintage Nintendo games for $2 per game.
The club charges a $20 annual membership.

Apply the rule of four to this scenario.

Graph:

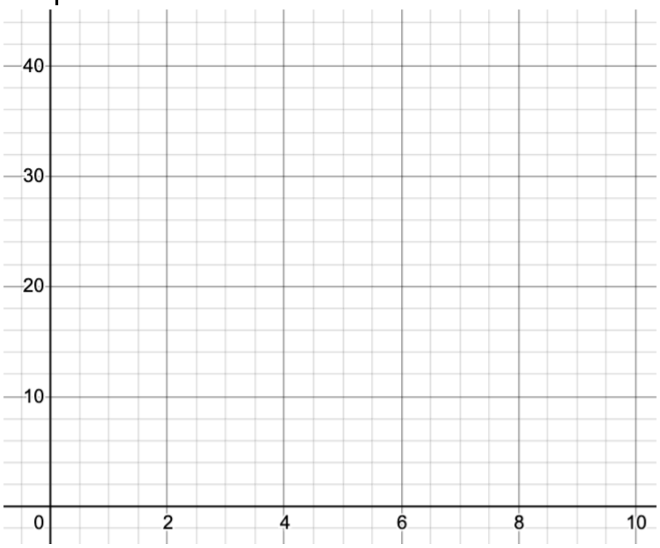


Table of Values

Number of Video Games Rented	Total Cost ($)
0	
2	
4	
6	
8	
10	

Words

Equation

Example 4

The equation, $C = 12n + 800$ describes the cost to rent a ballroom where C is the cost in dollars and n is the number of people.

Apply the Rule of Four to the equation.

Graph:

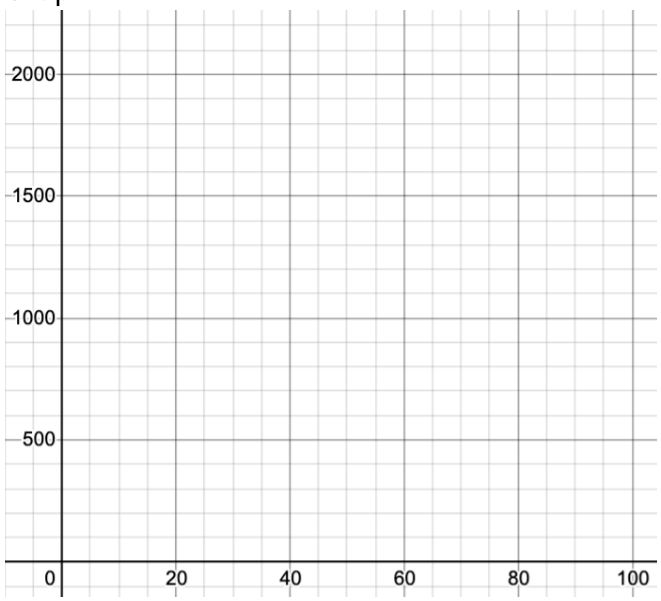


Table of Values

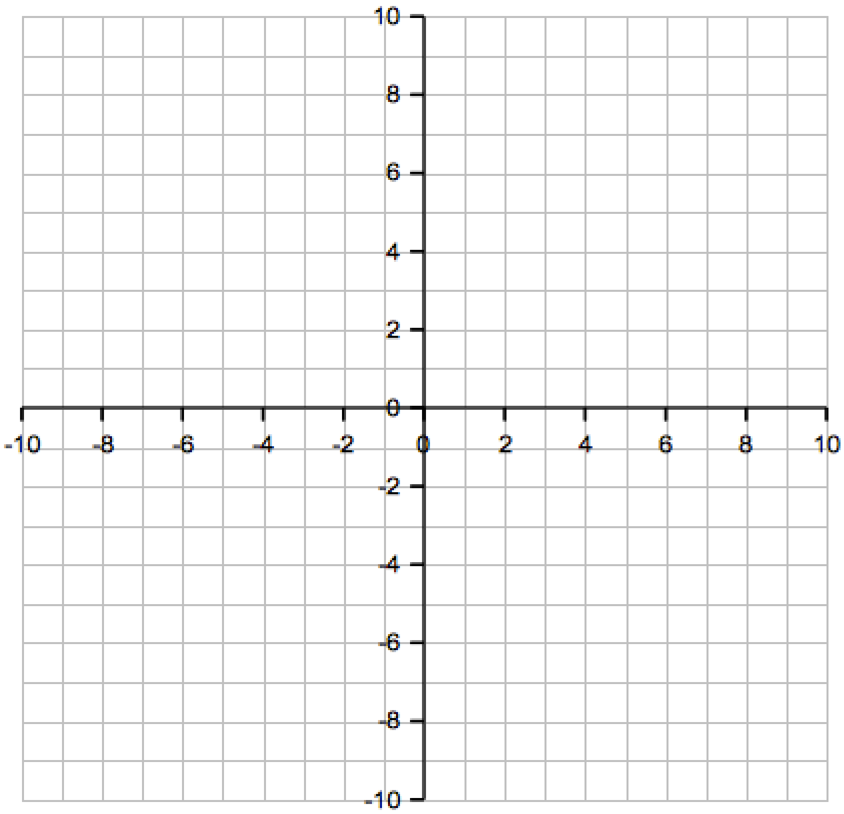
Number of People (n)	**Total Cost of Ballroom ($)**
0	
20	
40	
60	
80	
100	

Words

Equation

Example 5 – Apply the Rule of Four to the equation.

$$y = -\frac{2}{3}x + 5$$



Homework

[These are textbook questions]

4. Use the rule of four to represent this relation in three other ways.

a) Use a graph.

b) Use words.

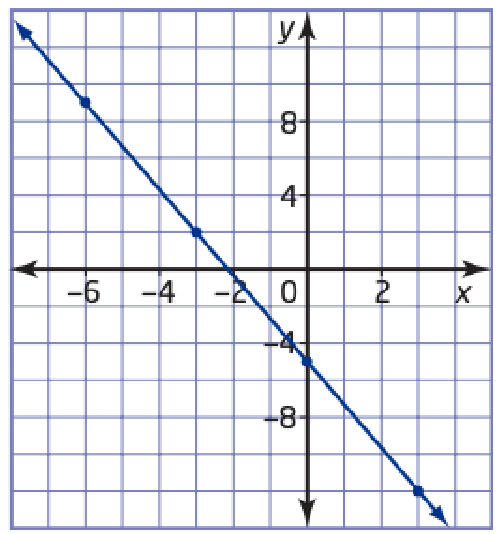
c) Use an equation.

x	y
0	2
1	5
2	8
3	11
4	14

8. The table shows how the depth of a scuba diver changes with time. Complete the rule of four for the relation by representing it using words, a graph, and an equation.

Time (s)	Depth (m)
0	–50
5	–45
10	–40
15	–35
20	–30

11. Complete the rule of four for this relation by representing it numerically, in words, and with an equation.



12. Complete the rule of four for the relation $y = 4x - 3$ by representing it numerically, graphically, and in words.